

ASSIGNMENT-1

Q.1 Conversion

(a) $(3d7e)_{16} = (\quad)_{10} = (\quad)_2$

Decimal

$$e = 14, d = 13, 7, 3$$

$$= (3 \times 16^3) + (13 \times 16^2) + (7 \times 16^1) + (14 \times 16^0)$$

$$= 12288 + 3328 + 112 + 14$$

$$= \boxed{(15742)_{10}}$$

Binary

$$3 = 10011$$

$$d = 1101$$

$$7 = 0111$$

$$e = 1110$$

$$= \boxed{(001111010111110)_2}$$

(b) $(764)_{10} = (\quad)_2 = (\quad)_{16}$

Binary

$$\begin{array}{r|l} 2 & 764 \end{array}$$

$$\begin{array}{r|l} 2 & 382 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 191 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 95 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 47 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 23 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 11 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 5 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 2 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & 1 \text{ @} \end{array}$$

$$\begin{array}{r|l} 2 & \end{array}$$

$$= \boxed{(1011111100)_2}$$

Hexadecimal

$$0010 = 2$$

$$1111 = f$$

$$1100 = c$$

$$= [(2fc)_{16}]$$

Q.2 - 39

a) 8-bit

$$\begin{array}{r|l} 2 & 39 \\ \hline 2 & 19 \text{ ①} \\ 2 & 9 \text{ ①} \\ 2 & 4 \text{ ①} \\ 2 & 2 \text{ ①} \\ & 1 \text{ ①} \end{array}$$

$$= [(100111)_2]$$

→ Padding to 8 bits

$$+39 = 00100111$$

→ Flip

$$\begin{array}{r} 11011000 \\ + \\ \hline 11011001 \end{array}$$

$$[-39 = 11100101]$$

(b) 16-bit

→ Padding to 16 bits

$$+39 = 0000000000100111$$

→ Flip

$$\begin{array}{r} 1111111111011000 \\ + \\ \hline 1111111111011001 \end{array}$$

(a)

Integer part

①

2

1

①

100

Normalize

• $\text{sign} = 0$

• Exponent = $-3 + 127 = 124$.

$$2/124$$

2 6 2 0

2 31 6

2, 18, ①

2	7	①
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2 3 ①

1 (1)

2 11111 00

$$= \underline{\underline{(01111100)_2}}$$

- Mantissa

001

pad to 23 bits

$$= (0 | 000000000000000000000000)$$

Answer

$\frac{2}{6} \text{ OIIIII } \text{OO} \quad \text{OIOOOOOOOOOOOOOOOOOOOOOO}$

(b)

22.7

2	1	
2	10	(1)
2	5	(2)
2	2	(1)
1	1	(0)

$$= 1011_2$$

001

0.1×2	0.2	0
0.2×2	0.4	0
0.4×2	0.8	0
0.8×2	1.6	1
0.6×2	1.2	1
0.2×2	0.4	—

loops
01 0001100
= 1011. ~~0001~~ ...

Answer Normalization

$1001100011001100 \dots \times 2^3$

- Exponent

$$127 + 3 = 130$$

130 in binary

$\approx 10000010_2$

- Mantissa

① 11000 1100 1100 1100 1100 10,

(i.e. 24 because
27th bit was
zero.)

Answer

0 1000000 01100011001100110010₂

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Q.4 DELL VOSTRO EXPRESS CODE

BQWMJ42

$$A = 10$$

$$B = 11$$

$$C = 12$$

.

.

.

$$Z = 35$$

Here,

$$B = 11$$

$$Q = 26$$

$$W = 32$$

$$M = 22$$

$$J = 19$$

$$4 = 4$$

$$2 = 2$$

$$= 11 \times 36^6 + 26 \times 36^5 + 32 \times 36^4 + 22 \times 36^3 + 19 \times 36^2 + 4 \times 36^1 + 2 \times 36^0$$

$$= \underline{(25,571,525,189)_{10}}$$

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Q.5

$$a = 1925$$

$$b = 3012$$

$$a) \quad c = a + b$$

$$= 1925 + 3012$$

$$= 4937$$

Crossed 4095 (overflow)

$$c = 4937 - 4096$$

$$\boxed{c = 841}$$

$$b) \quad c = a - b$$

$$= 1925 - 3012$$

$$= -1087$$

$$\neq c = 4096 - 1087$$

(underflow)

$$\boxed{c = 3009}$$

$$c) \quad a + b \quad (\text{Two's complement (2 bit)})$$

$$a = 1925 \quad (\text{within positive range } (< 2048))$$

$$b = 3012 \quad (\text{stay same) (outside the range)}$$

$$b = 3012 - 4096$$

$$\neq c = 1925 - 1084$$

$$\boxed{c = 841}$$

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Q.6

Original (Hex)

Truncated (Hex)

0 (00000)

0 (0000)

3 (00011)

3 (0011)

11 (10001)

1 (0001)

17 (10111)

7 (0111)

1F (11111)

F (1111)

Original (Unsigned)

Truncated (Unsigned)

0

0

3

3

17

7

23

7

31

15

Two's Comp Original

Two's comp Truncated

0

0

3

3

-15

1

-9

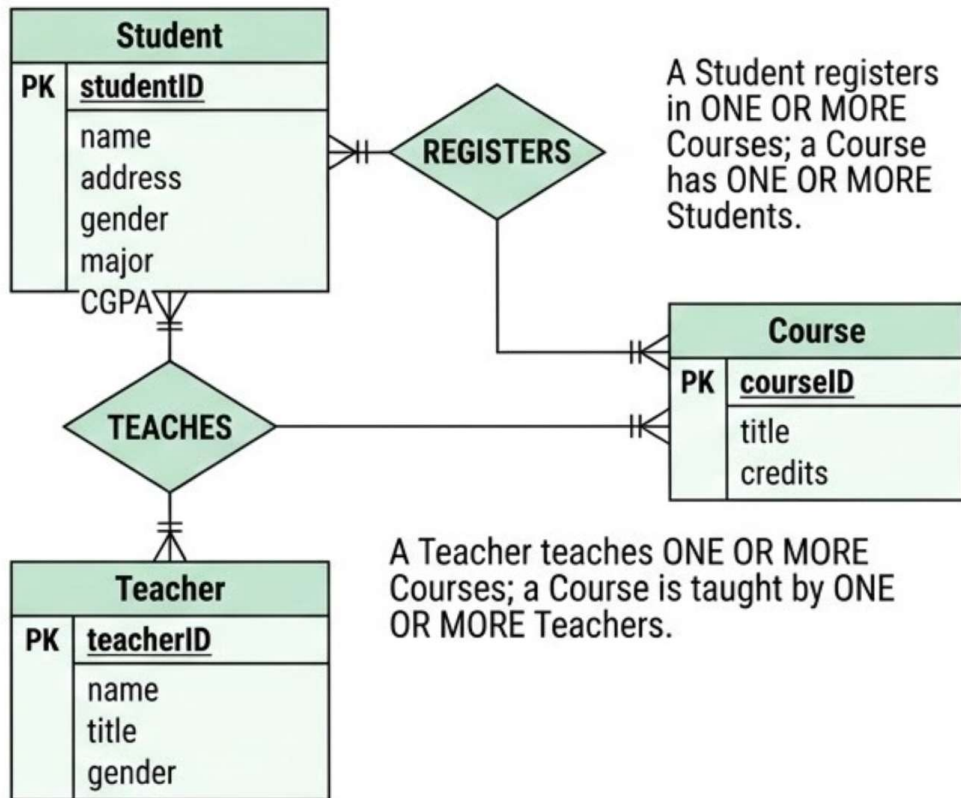
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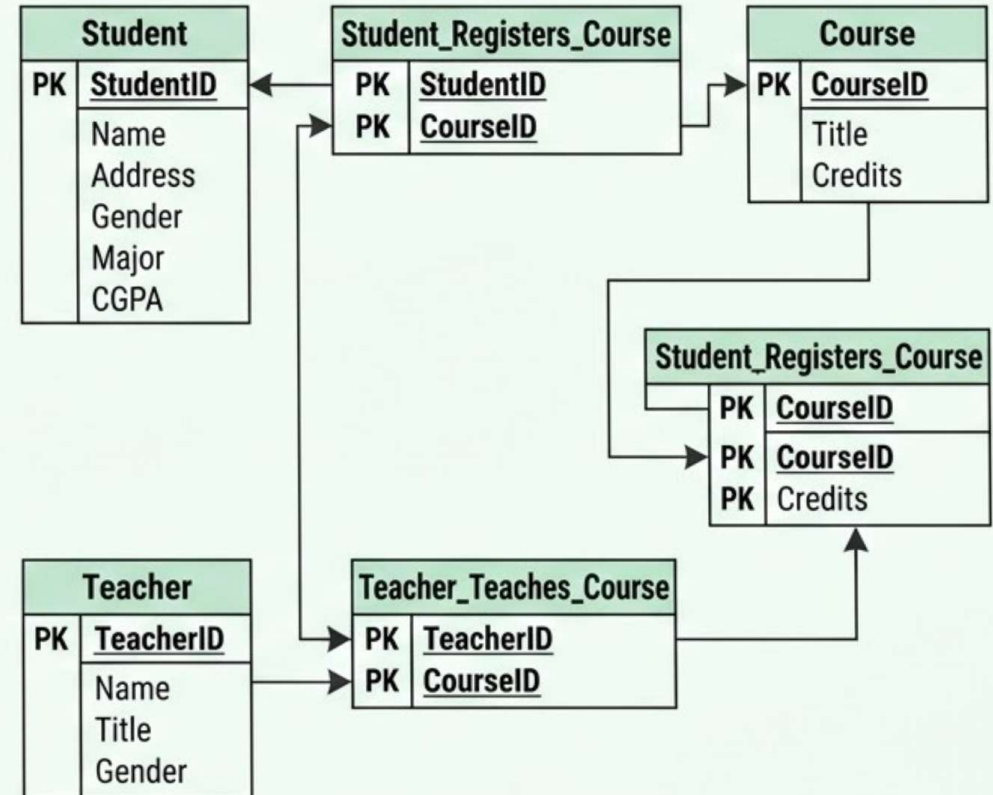
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College Database Information System Diagrams

ENTITY-RELATIONSHIP DIAGRAM (COLLEGE DATABASE)

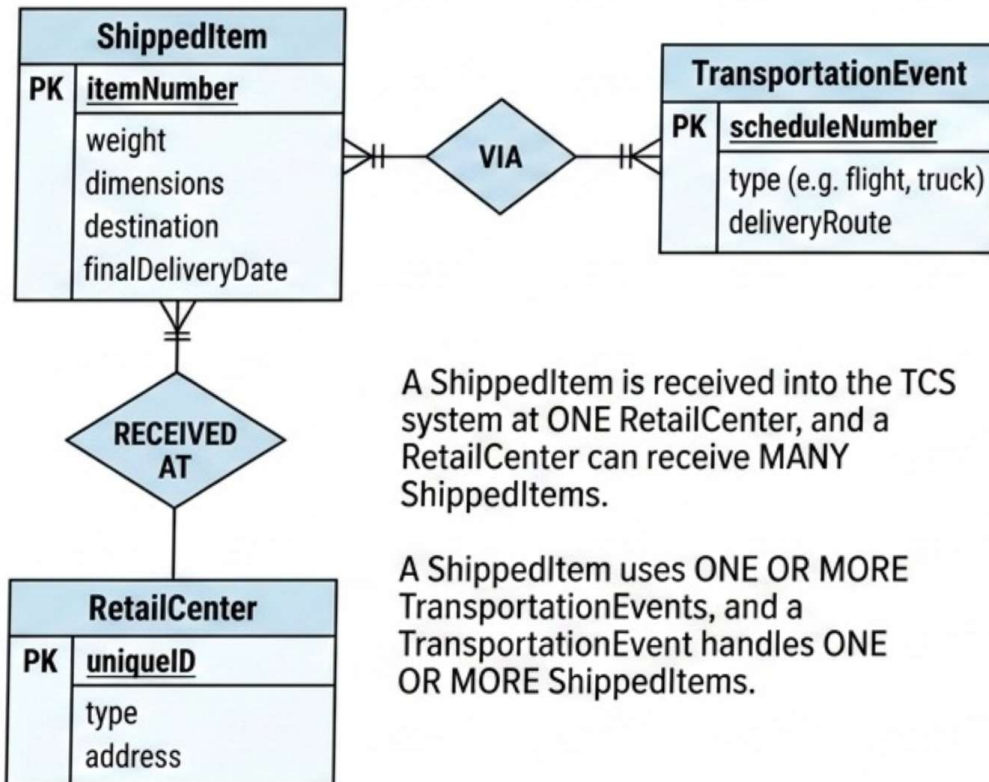


CORRESPONDING RELATIONAL MODEL (SCHEMA)



TCS Shipping Information System Diagrams

ENTITY-RELATIONSHIP DIAGRAM (TCS SHIPPING)



CORRESPONDING RELATIONAL MODEL (SCHEMA)

